

WEEK 5 REPORT

PENTAS HUB Internship Program

Project Title: Hybrid AI-Based Diabetes and Heart Disease Risk Prediction System with RAG-Based Medical Explanation Chatbot

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Student ID: Your Student ID

Week: 5

Domain: Machine Learning & Artificial Intelligence

1. Objective

Develop a RAG-based medical explanation chatbot, integrate it with the disease prediction system, and deploy the complete application using Streamlit.

2. Tasks Assigned

Develop RAG chatbot; create medical knowledge base; implement TF-IDF retrieval; generate explanations; integrate chatbot with prediction models; deploy using Streamlit; prepare documentation.

3. Tasks Completed

Successfully developed the RAG chatbot, created the knowledge base, implemented retrieval, integrated hybrid ML and BERT models, and built a Streamlit web application.

4. Technologies Used

Python, Scikit-learn, Streamlit, Hugging Face Transformers, PyTorch, Pandas, NumPy, Joblib, TF-IDF Vectorizer, Cosine Similarity, GitHub.

5. Files Developed

Datasets: diabetes_cleaned.csv, heart_cleaned.csv, patient_notes.csv, medical_text_dataset.csv, medical_knowledge_base.txt.

Models: hybrid_diabetes_model.pkl, hybrid_heart_model.pkl, bert_model/.

Chatbot: rag_chatbot.py, knowledge_base.py, retrieve_documents.py, explain_prediction.py, prompt_template.py.

Web App: app.py, prediction.py, chatbot_ui.py.

6. Methodology

Loaded trained models, loaded BERT, created knowledge base, generated TF-IDF vectors, retrieved relevant documents using cosine similarity, generated explanations, integrated modules, and deployed with Streamlit.

7. Results

Successfully predicted diabetes risk, heart disease risk, classified medical text, generated RAG explanations, and deployed the final application.

8. Challenges Faced

Understanding RAG, integrating multiple modules, organizing the knowledge base, and deploying the application.

9. Solutions Implemented

Used TF-IDF retrieval, cosine similarity, modular programming, and Streamlit for deployment.

10. Learning Outcomes

Learned RAG concepts, transformer-based NLP, information retrieval, model integration, and web deployment.

11. Conclusion

Successfully completed the final AI healthcare prediction system by integrating hybrid machine learning models, BERT-based medical text classification, a RAG-based explanation chatbot, and a Streamlit web interface into a single application.